

1. Use a calculator or spreadsheet to find $S(t)$ for the values of t given in the following table.

t	$S(t) = 5.3t - 14.8$
11	
17	
25	
46	

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11	
17	
25	
46	

2. Suppose $f(x) = x^2 - 9x$ million dollars are earned, where x is the number of years after 1990.

- a. What is $f(10)$?
 b. The answer to part (a) gives the number of millions of dollars earned for what year?

- a. What is $f(10)$?

$f(10) =$

- b. The answer to part (a) gives the number of millions of dollars earned for what year?

3. Using data starting with 1950 and projected to 2050, the number of women (in millions) in the workforce is given by the function $P = 0.79x + 20.86$, where x equals the number of years after 1950.

- (a) Find the value of P when $x = 66$. Explain what this means.
 (b) What value of x represents 2040? Use the model to find the number of women in the workforce in 2040.

- (a) The value of P when $x = 66$ is million.
 (Type an integer or decimal rounded to two decimal places as needed.)

What is the meaning of this value of P ?

- ☐ A. The value of P represents the number of women in the workforce in the year 1966.
☐ B. The value of P represents the number of women who will join the work force between the years 1900 and 1966.
☐ C. The value of P represents the number of women who will join the work force between the years 1950 and 2016.
☐ D. The value of P represents the number of women in the workforce in the year 2016.

- (b) The value of x that represents 2040 is $x =$.

The number of women in the workforce in 2040 will be million.
 (Type an integer or decimal rounded to two decimal places as needed.)

4. The amount of federal tax per capita (per person) can be modeled by the function $S = 3.3267t^3 - 56.48t^2 + 549.85t + 5158.9$ where t is the number of years after 1990.
- a. What are the values of t that correspond to 1997, 2017, and 2020?
- b. $S = f(17)$ gives the value of S for what year? What is $f(17)$?
- c. What $x_{\min}$ and $x_{\max}$ should be used to set a viewing window so that t represents 1990–2020?
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a. What value of t corresponds to the year 1997?

$t =$

What value of t corresponds to the year 2017?

$t =$

What value of t corresponds to the year 2020?

$t =$

b. $S = f(17)$ gives the value of S for what year?

What is $f(17)$?

$f(17) =$

(Round to the nearest integer as needed.)

c. What $x_{\min}$ should be used for the viewing window?

$x_{\min} =$

What $x_{\max}$ should be used for the viewing window?

$x_{\max} =$

- *5. Using a graphing calculator, estimate the real zeros, the relative maxima and minima, and the range of the polynomial function.

$$g(x) = x^3 - 1.2x + 3$$

The zero(s) of the function is/are at approximately $x =$.

(Round to three decimal places as needed. Use a comma to separate answers as needed.)

Find the relative maxima. Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

☐ A. The function has a relative maximum of at $x \approx$.

(Round to three decimal places as needed.)

☐ B. The function does not have any relative maxima.

Find the relative minima. Select the correct choice below and, if necessary, fill in the answer box to complete your choice.

☐ A. The function has a relative minimum of at $x \approx$.

(Round to three decimal places as needed.)

☐ B. The function does not have any relative minima.

The range of the function is .

(Type your answer in interval notation.)